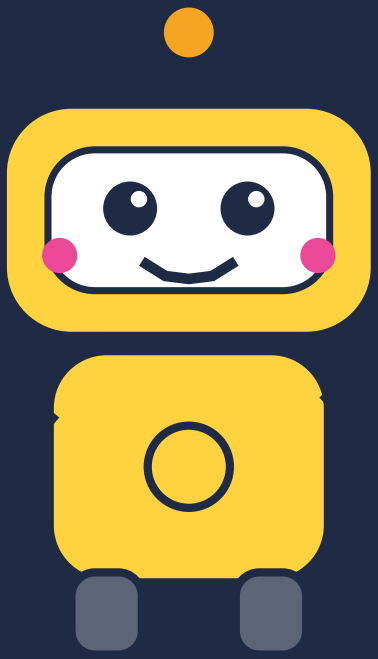
TEACHER BOOK

Think Like a Coder

Teacher book for Class 6

Period-by-period plans for every CT and AI chapter
Teacher guides for both 20-hour projects
Answer keys, assessment guide and photocopiable journal



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 6 · Follows NCERT Class 6 maths, CBSE's four AI chapters and two projects

SAMPLE EDITION · DRAFT

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About this teacher book

This book goes with the *Think Like a Coder* Class 6 workbook. It gives you a plan for every chapter, so that a maths or class teacher can teach computational thinking and AI confidently without being a computer specialist. CBSE suggests about **100 hours** a year for Class 6; the plans here use **150 periods of 40 minutes**, which is about 100 hours.

★ What is in this book

- ✓ **A year plan** showing which chapter to teach in which term, next to the NCERT maths chapter.
- ✓ **A period-by-period plan for every chapter**, including the AI chapters: what you do and what students do.
- ✓ **A hands-on activity** for every chapter, with set-up steps and a tip.
- ✓ **Common mistakes** to watch for, and how to support or stretch students.
- ✓ **An exit check** with answers, and what to note in the observation journal.
- ✓ **The answer key** for every page of the student workbook.
- ✓ **Photocopiable** observation journal and term summary pages at the back.
- ✓ **A teacher guide for each 20-hour project**: week-by-week plan, roles, set-up, pitfalls and how to mark with the rubric.

✓ Before you start

- ✓ Read the chapter in the student book and these notes. Most lessons need 5 to 10 minutes to prepare.
- ✓ Get the kit items ready (listed under *At a glance*). Everything is reusable.
- ✓ Plan pairs in advance: mix confident readers with children who prefer talking and acting out.
- ✓ Teach the CT chapter in the same week or fortnight as the matching NCERT maths chapter, so the maths is fresh.
- ✓ Keep it active: children should talk, act out and build more than they listen.

Computers are optional

No computer lab is needed. Every CT and AI idea is taught with paper, cards and talk, and both projects run on paper. Where computers are available, the optional boxes in the AI chapters and projects suggest short demonstrations. A projector helps but is optional.

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How a lesson runs

Every lesson fits one 40-minute period and follows the same five steps, based on the PRIMM approach used in computing classrooms (predict, run, investigate, modify, make). Children learn the routine once and then focus on the thinking.

1. **Predict and run** 8 min
   Children predict what a set of steps will do, then act it out.
2. **Investigate** 8 min
   Pairs answer 'why' questions about how it worked.
3. **Modify** 8 min
   Children change the steps to do something new.
4. **Put in order** 8 min
   Jumbled code cards go back in the right order.
5. **Make** 8 min
   Pairs write their own program and swap with another pair.

How it supports CBSE's assessment

CBSE asks schools to assess CT&AI through written puzzles, group activities and a **Teacher Observation Journal**. Each step gives the teacher something to observe: predictions show reasoning, modifications show understanding, and the 'make' step shows whether a child can build an algorithm of their own. The journal format and a term summary for the report card come with the teacher book.

Every child can take part

- ✓ Children who finish early move to the challenge level; children who need help use a hint card.
- ✓ Most activities are done in pairs, so children who find reading hard can still take part by talking and acting out.
- ✓ Teacher preparation is about 5-10 minutes per lesson using the script.

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Year plan

10 CT chapters of 6 periods (about 40 hours), 4 AI chapters of 7 periods plus 2 review periods (about 20 hours), and two projects of 30 periods each (about 40 hours): **150 periods** of 40 minutes in all. Teach each chapter alongside the NCERT maths chapter with the same name; the order and timing can follow your school's maths plan.

Ch	Chapter	Coder skill	Level	Periods	Term
1	Patterns in Mathematics	Rules and functions: name a rule and reuse it	C1	6	Term 1
2	Lines and Angles	Drawing programs: angles, turns and a square(size) function	C1	6	Term 1
3	Number Play	Algorithms that repeat until something happens	C1	6	Term 1
4	Data Handling and Presentation	Lists: store, sort and find the biggest	C1	6	Term 1
5	Prime Time	The sieve and a factor-finding algorithm	C2	6	Term 1
6	Perimeter and Area	Functions with two inputs; breaking shapes apart	C2	6	Term 2
7	Fractions	Compare, simplify and sort fractions	C2	6	Term 2
8	Playing with Constructions	Construction scripts: exact steps and debugging	C2	6	Term 2
9	Symmetry	Reflect and rotate: functions that change shapes	C2	6	Term 2
10	The Other Side of Zero	Integers in data: lowest, highest and range	C2	6	Term 2
AI1	Introduction to AI	Tell AI from automation; how machines learn	C3	7	Term 1
AI2	Basic Data Concepts	Types of data; collect, organise and show it	C3	7	Term 1
AI3	Patterns and Decisions	Learn a rule from labelled examples, then decide	C3	7	Term 2
AI4	Ethics and Digital Responsibility	Strong passwords, privacy and digital footprints	C3	7	Term 2
P1	Weather Across India	Collect, organise and present real data	C3	30	Term 1
P2	A Smarter School Routine	Design, test and improve an algorithm	C3	30	Term 2
	Revision and term assessment	Review games, the term summary and parent feedback	All	2	Both

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Assessment and the observation journal

CBSE asks for continuous, competency-based assessment in CT&AI: written puzzles, group activities and a **Teacher Observation Journal**. This book makes that simple. In each chapter, the exit check and the 'what to note' box tell you what to look for; record one level per child per skill every few chapters, not every lesson.

Skill	Beginning	Developing	Secure	Extending
Algorithms and functions	Follows given steps	Writes steps with loops	Writes and reuses a function	Designs and tests functions with inputs
Data	Reads a table	Organises data in a table or list	Charts data; finds lowest, highest, range	Draws fair conclusions and states limits
AI ideas	Names AI examples	Tells AI from automation	Explains learning from labelled data	Explains bias and how better data helps
Projects	Takes part with help	Completes stages with help	Plans, tests and presents with the team	Improves the work using evidence

How to assess without extra tests

- **Watch and listen** during activities: explaining, predicting and fixing show understanding better than a written answer.
- **Use the exit check** at the end of each chapter: two quick questions, answers in this book.
- **Compare with self-checks**: each chapter ends with 'How did I do?', where children colour a face for each 'I can' statement.
- **Summarise each term** on the term summary page: one level per area and a short comment for the report card.

The four levels

Use the four levels for every skill. **Beginning**: needs help to start. **Developing**: succeeds with prompts or a partner. **Secure**: succeeds alone on familiar problems. **Extending**: solves new problems and explains why. Most children will be Developing or Secure; that is expected and healthy.

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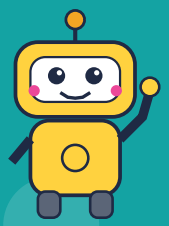
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TEACHER NOTES · CHAPTER 1 · LEVEL C1

Patterns in Mathematics

Coder skill: Rules and functions: name a rule and reuse it

Goes with NCERT maths, Class 6, chapter 1



At a glance	
Students will	describe a sequence with a rule, use a function with an input and spot square and triangular numbers.
Key words	**sequence** (numbers that follow a rule), **term** (one number in a sequence), **function** (a named rule with an input), **output** (what the function gives back)
Materials	Counters for dot patterns, function cards (square, triangle, f(n) = 3 x n - 1), small whiteboards
Time	6 periods of 40 minutes
Link	Teach alongside chapter 1 of the NCERT Class 6 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *sequence*, *term*, *function*, *output*. Work through the example *Adding odd numbers makes squares* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Function machine relay	Run the activity below. Move between groups and note who can explain the idea.	Call and check functions with inputs, build square numbers from odd layers, and invent their own function.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a function called **triangle(n)** that adds 1 + 2 + ... + n and gives back the total. Model the first card or step, then let pairs finish.	Run triangle(4) and triangle(6) by hand. What does each give back?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Function machine relay

1. Write three function cards on the board: *square(n)*, *triangle(n)* and *f(n) = 3 x n - 1*.
2. Teams take turns: one student calls a function with an input, the next gives the output, the next checks it.
3. Groups build square numbers with counters, adding each odd number as an L-shaped layer (1, 3, 5, 7).
4. Ask: 'Why give a rule a name?' Link to reusing a function instead of repeating the same steps.

Tip: Let students invent a function and its name. Naming a rule is the step from pattern to program.

Common mistakes to watch for

- **Confusing 'next term' rules with 'position' rules.** Ask which one lets you jump to the 100th term.
- **Off-by-one errors** in the position rule: the 10th term of 5, 9, 13 is 5 + 9 x 4.
- **Treating square(n) as 2 x n.** Square means n x n.

Different paces

Support: Use only add-on rules and small inputs, with counters.
Stretch: Find a function for 2, 6, 12, 20, 30 (n x (n + 1)) and test it on the 10th term.

Exit check and what to note

Two quick questions at the end of the chapter:

1. What is square(7)? (Answer: 49)

2. What is triangle(5), that is 1 + 2 + 3 + 4 + 5? (Answer: 15)

Note in the journal: Beginning (needs help to describe a sequence with a rule); **Developing** (can describe a sequence with a rule with prompts or a partner); **Secure** (can describe a sequence with a rule and use a function with an input alone); **Extending** (can spot square and triangular numbers and explain why).

Answer key

1. 36 and 49
2. 21 (add 6)
3. a) 41 (5 + 9 x 4)
4. f(4) = 11; n = 10
5. b) 6
Build it: triangle(4) = 10; triangle(6) = 21
Challenge: 400: the first n odd numbers add up to n x n, and 20 x 20 = 400

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TEACHER NOTES · CHAPTER 2 · LEVEL C1

Lines and Angles

Coder skill: Drawing programs: angles, turns and a square(size) function

Goes with NCERT maths, Class 6, chapter 2



At a glance	
Students will	name and measure angles, turn on an 8-point compass and write and reuse a drawing function.
Key words	**angle** (an amount of turn), **acute** (less than 90°), **obtuse** (between 90° and 180°), **reflex** (more than 180°)
Materials	Compass-point signs for the classroom walls (N, NE, E, SE, S, SW, W, NW), squared paper, protractors or turn cards
Time	6 periods of 40 minutes
Link	Teach alongside chapter 2 of the NCERT Class 6 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *angle*, *acute*, *obtuse*, *reflex*. Work through the example *Turning on the 8-point compass* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Compass drill and square(size)	Run the activity below. Move between groups and note who can explain the idea.	Turn on an 8-point compass, then draw and reuse shape functions with the right turns.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a **square(size)** function and use it twice. Model the first card or step, then let pairs finish.	Now change it into **triangle(size)** for a triangle with equal sides. What goes in the repeat and the turn?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Compass drill and square(size)

1. Put the 8 compass-point signs on the walls. Everyone stands and faces North.
2. Call turns: '45° clockwise', '135° anticlockwise'. Students turn and name the new direction.
3. On squared paper, pairs follow *square(4)* and *square(2)* drawn from the same corner.
4. Pairs work out the turn for a triangle and a hexagon (360 ÷ sides) and write *triangle(size)*.

Tip: Turning the body first, then drawing, helps students who mix up clockwise and anticlockwise.

Common mistakes to watch for

- **Using the inside angle as the turn**: a triangle's turn is 120°, not 60°.
- **Each compass step is 45°**, not 90°: NE is between N and E.
- **Calling an acute angle obtuse**: obtuse is between 90° and 180°.

Different paces

Support: Use the 4 main directions and 90° turns only.
Stretch: Write *polygon(sides, size)* that draws any regular shape, using the turn 360 ÷ sides.

Exit check and what to note

Two quick questions at the end of the chapter:

1. You face South-East and turn 90° anticlockwise. Which way do you face? (Answer: North-East)

2. What turn does a turtle make at each corner of a regular pentagon? (Answer: 72°)

Note in the journal: Beginning (needs help to name and measure angles); **Developing** (can name and measure angles with prompts or a partner); **Secure** (can name and measure angles and turn on an 8-point compass alone); **Extending** (can write and reuse a drawing function and explain why).

Answer key

1. c) obtuse (120°)
2. a) North-West
3. 120° (4 x 30°)
4. 60
5. b) 45°
Build it: Repeat 3 times: forward size, turn right 120° (3 x 120° = 360°)
Challenge: Repeat 4 times: square(30), turn right 90° (each square starts at the centre)

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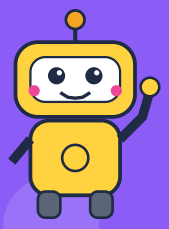
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TEACHER NOTES · CHAPTER 3 · LEVEL C1

Number Play

Coder skill: Algorithms that repeat until something happens

Goes with NCERT maths, Class 6, chapter 3



At a glance	
Students will	follow and trace an algorithm, use 'repeat until' and explore number puzzles like palindromes.
Key words	**algorithm** (exact steps to solve a problem), **repeat until** (loop that stops when something is true), **palindrome** (reads the same both ways), **trace** (follow steps and write each value)
Materials	Trace tables, calculators (optional, for checking), algorithm cards (reverse-and-add, Kaprekar, halve-or-3n+1)
Time	6 periods of 40 minutes
Link	Teach alongside chapter 3 of the NCERT Class 6 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *algorithm*, *repeat until*, *palindrome*, *trace*. Work through the example *Reverse and add, starting with 57* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Algorithm race	Run the activity below. Move between groups and note who can explain the idea.	Trace repeat-until algorithms on different inputs, count steps, and compare results.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Kaprekar's routine for 3-digit numbers (digits not all the same). Model the first card or step, then let pairs finish.	Trace it for 352. Write each subtraction. How many steps does it take?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Algorithm race

1. Give groups different start numbers for reverse-and-add (39, 57, 68, 89).
2. Each group traces the algorithm in a table until it reaches a palindrome, counting steps.
3. Compare: which start took longest? (89 takes many steps: leave it as a stretch.)
4. Repeat with Kaprekar's routine for 3-digit numbers and see every group reach 495.

Tip: Tell groups that 89 takes 24 steps before starting it. They can decide whether to 'run' it by hand.

Common mistakes to watch for

- **Stopping too early** before checking that the number really is a palindrome.
- **Dropping a leading zero** in Kaprekar's routine: 99 must be written 099.
- **Arithmetic slips** in long traces. Partners check each line.

Different paces

Support: Use reverse-and-add with 2-digit starts that finish in one or two steps.
Stretch: Find a 2-digit start that takes exactly 3 reverse-and-add steps (for example 59: 154, 605, 1111).

Exit check and what to note

Two quick questions at the end of the chapter:

1. Reverse and add, starting with 48. How many steps to a palindrome, and which palindrome? (Answer: 2 steps: 132, then 363)

2. Halve if even, else x 3 + 1. Starting at 5, how many steps to reach 1? (Answer: 5 (16, 8, 4, 2, 1))

Note in the journal: Beginning (needs help to follow and trace an algorithm); **Developing** (can follow and trace an algorithm with prompts or a partner); **Secure** (can follow and trace an algorithm and use 'repeat until' alone); **Extending** (can explore number puzzles like palindromes and explain why).

Answer key

1. b) 4554
2. 2 steps: 39 + 93 = 132, 132 + 231 = 363
3. 47 and 59
4. 8 steps: 6, 3, 10, 5, 16, 8, 4, 2, 1
5. b) 4: two 6s make 12; the other 8 needs at least 2 rolls with no 6 (such as 4 + 4)
Build it: 532 - 235 = 297, 972 - 279 = 693, 963 - 369 = 594, 954 - 459 = 495: 4 steps
Challenge: 211 - 112 = 099, 990 - 099 = 891, 981 - 189 = 792, 972 - 279 = 693, 963 - 369 = 594, 954 - 459 = 495: it still reaches 495, in 6 steps

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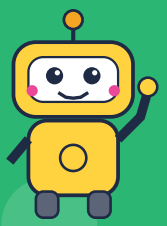
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TEACHER NOTES · CHAPTER 4 · LEVEL C1

Data Handling and Presentation

Coder skill: Lists: store, sort and find the biggest

Goes with NCERT maths, Class 6, chapter 4



At a glance	
Students will	store data in a list, read pictographs and bar graphs with a scale and find the biggest item with an algorithm.
Key words	**data** (facts collected to answer a question), **list** (items in order, by position), **frequency** (how often something occurs), **scale** (what one unit stands for)
Materials	Large number cards for a 'human list', survey sheets, squared paper for bar graphs
Time	6 periods of 40 minutes
Link	Teach alongside chapter 4 of the NCERT Class 6 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *data*, *list*, *frequency*, *scale*. Work through the example *Favourite sports of 30 students* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The human list	Run the activity below. Move between groups and note who can explain the idea.	Act out a list and the 'biggest so far' algorithm, then collect survey data and graph it with a scale.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	An algorithm that finds the biggest item in a list. Model the first card or step, then let pairs finish.	Trace it on the list 12, 19, 7, 25, 14. Write the value of biggest after each item.
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: The human list

1. Six students stand in a row holding value cards; their positions are 1 to 6.
2. Run 'find the biggest': a 'biggest so far' card starts with student 1 and is passed along only when a bigger value appears.
3. Ask: 'How many comparisons did we make?' (5 for 6 items.)
4. Groups run a quick survey, record the answers as a list, then draw a bar graph with a sensible scale.

Tip: Ask 'What is in position 3?' and 'In which position is the 21?'. Students must tell position from value.

Common mistakes to watch for

- **Mixing up position and value** in a list.
- **Bar graphs without a scale**, or with uneven steps on the axis.
- **Pictograph keys ignored**: one symbol may stand for 4 or 5 items.

Different paces

Support: Use a list of 4 numbers and a graph where 1 square = 1.
Stretch: Change the algorithm to find both the biggest and the smallest in one pass.

Exit check and what to note

Two quick questions at the end of the chapter:

1. In the list [14, 9, 21, 7], what is in position 3? (Answer: 21)

2. In a bar graph, 1 square = 4 students. A bar is 6 squares tall. How many students? (Answer: 24)

Note in the journal: Beginning (needs help to store data in a list); **Developing** (can store data in a list with prompts or a partner); **Secure** (can store data in a list and read pictographs and bar graphs with a scale alone); **Extending** (can find the biggest item with an algorithm and explain why).

Answer key

1. 33 and 35
2. 6 (16 - 10)
3. 35
4. b) 23
5. b) 16 (6 x 16 = 96; 6 x 17 = 102 is too many)
Build it: 12, 19, 19, 25, 25: it says 25
Challenge: Use 'smallest' instead of 'biggest' and 'If item < smallest'; the shortest is 128 cm

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5

TEACHER NOTES · CHAPTER 5 · LEVEL C2

Prime Time

Coder skill: The sieve and a factor-finding algorithm

Goes with NCERT maths, Class 6, chapter 5



At a glance	
Students will	find factors and multiples, use the sieve to find primes and explain a factor-finding algorithm.
Key words	**factor** (divides a number exactly), **multiple** (a number in the times table), **prime** (exactly two factors), **sieve** (an algorithm that filters)
Materials	1-100 number grids, coloured pencils, string or wool, a cardboard circle with 12 points (or a clock face)
Time	6 periods of 40 minutes
Link	Teach alongside chapter 5 of the NCERT Class 6 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *factor*, *multiple*, *prime*, *sieve*. Work through the example *The sieve for 1 to 30* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Sieve and string art	Run the activity below. Move between groups and note who can explain the idea.	Run the sieve to find primes to 100, and explore jumps on a 12-point circle using factors.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	An algorithm that lists the factors of n and checks if n is prime. Model the first card or step, then let pairs finish.	Run it for n = 15 and n = 13. What is in the list each time, and does it say 'prime'?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Sieve and string art

1. Pairs run the sieve on a 1-100 grid: cross out 1, circle 2 and cross its multiples, then 3, 5 and 7.
2. Ask: 'Why can we stop after 7?' (The next prime is 11, and 11 x 11 = 121 is more than 100, so every non-prime up to 100 has a factor of 7 or less.)
3. On a 12-point circle, wind string from point to point in jumps of 2, 3, 4 and 5.
4. Record how many points each jump visits, and link it to the factors of 12.

Tip: Use different colours for each prime's multiples. Numbers crossed twice show common multiples.

Common mistakes to watch for

- **Calling 1 a prime.** A prime has exactly two factors.
- **Forgetting a prime's own place**: circle 7, do not cross it out.
- **Thinking every odd number is prime**: 9, 15 and 21 are not.

Different paces

Support: Run the sieve for 1-50 with 2, 3 and 5 only.
Stretch: Predict, then check, how many points jumps of 1 to 11 visit on the 12-point circle, using 12 ÷ (the common factor).

Exit check and what to note

Two quick questions at the end of the chapter:

1. How many primes are there between 30 and 50? (Answer: 5: 31, 37, 41, 43, 47)

2. On a 12-point circle, jumping 4 at a time, how many different points are visited? (Answer: 3)

Note in the journal: Beginning (needs help to find factors and multiples); **Developing** (can find factors and multiples with prompts or a partner); **Secure** (can find factors and multiples and use the sieve to find primes alone); **Extending** (can explain a factor-finding algorithm and explain why).

Answer key

1. 10
2. c) 29
3. d) 6
4. d) 12 (5 and 12 share no factor except 1, so every point is visited)
5. 9: 1, 2, 3, 4, 6, 9, 12, 18, 36
Build it: 15: [1, 3, 5, 15], not prime; 13: [1, 13], prime
Challenge: 1, 5, 7 and 11: they share no factor with 12 except 1

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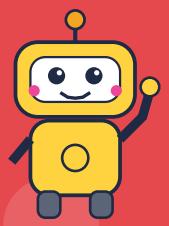
6

TEACHER NOTES · CHAPTER 6 · LEVEL C2

Perimeter and Area

Coder skill: Functions with two inputs; breaking shapes apart

Goes with NCERT maths, Class 6, chapter 6



At a glance	
Students will	find perimeter and area, use a function with two inputs and break a shape into rectangles.
Key words	**perimeter** (the distance around), **area** (the space inside), **decompose** (break into smaller parts), **input** (a value given to a function)
Materials	Metre sticks or measuring tapes, squared paper, a floor plan of the classroom or school
Time	6 periods of 40 minutes
Link	Teach alongside chapter 6 of the NCERT Class 6 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *perimeter*, *area*, *decompose*, *input*. Work through the example *The area and perimeter of an L-shaped garden* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Floor-plan surveyors	Run the activity below. Move between groups and note who can explain the idea.	Measure real spaces, use area and perimeter functions, and decompose L-shapes in two ways.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Use one function inside another. Model the first card or step, then let pairs finish.	A classroom is 5 m by 4 m and tiles cost Rs 60 per square metre. What does the program say?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Floor-plan surveyors

1. Groups measure a rectangular space (a table, a door, the classroom) and record length and width.
2. They call the functions *area(l, w)* and *perimeter(l, w)* for each measurement.
3. On squared paper, draw an L-shaped room and split it into two rectangles in two different ways.
4. Check that both ways give the same area.

Tip: Ask groups to split the L-shape differently from their neighbours. Same answer, different decomposition.

Common mistakes to watch for

- **Adding instead of multiplying** for area, or forgetting to double for perimeter.
- **Missing the hidden side** of an L-shape when finding its perimeter.
- **Mixing units**: m and cm in the same calculation.

Different paces

Support: Use whole-metre rectangles on squared paper before measuring.
Stretch: Find the cost of tiling the classroom with *floor_cost(l, w, rate)* at Rs 60 per square metre.

Exit check and what to note

Two quick questions at the end of the chapter:

1. What are area(9, 4) and the perimeter of a 9 by 4 rectangle? (Answer: 36 and 26)

2. An L-shape is a 6 by 2 rectangle with a 2 by 3 rectangle on top. What is its area? (Answer: 18 (12 + 6))

Note in the journal: Beginning (needs help to find perimeter and area); **Developing** (can find perimeter and area with prompts or a partner); **Secure** (can find perimeter and area and use a function with two inputs alone); **Extending** (can break a shape into rectangles and explain why).

Answer key

1. 35 cm^2 and 24 cm
2. b) 25 cm^2 (side 5 cm)
3. b) 12 by 2 (28 cm, against 20 cm)
4. 40 m^2 (10 x 2 + 4 x 5)
5. d) each triangle has an area of 8 cm^2, not 16
Build it: area(5, 4) = 20; 20 x 60 = Rs 1,200
Challenge: Shortest: a 3 x 4 rectangle (14 units). Longest: a 1 x 12 strip, or any shape whose squares never form a 2 x 2 block (26 units)

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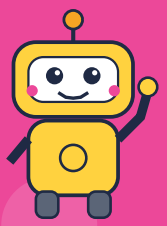
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TEACHER NOTES · CHAPTER 7 · LEVEL C2

Fractions

Coder skill: Compare, simplify and sort fractions

Goes with NCERT maths, Class 6, chapter 7



At a glance	
Students will	simplify fractions, compare and order fractions and trace a sorting pass.
Key words	**equivalent** (the same amount), **simplest form** (cannot be simplified more), **compare** (find which is bigger), **bubble sort** (swap neighbours until sorted)
Materials	Fraction cards for students to hold, fraction strips or circles, mark-sheet examples
Time	6 periods of 40 minutes
Link	Teach alongside chapter 7 of the NCERT Class 6 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *equivalent*, *simplest form*, *compare*, *bubble sort*. Work through the example *Which subject did Riya do best in?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Fraction bubble sort	Run the activity below. Move between groups and note who can explain the idea.	Sort fractions with compare-and-swap, explain each comparison, and simplify fractions step by step.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	An algorithm that puts a fraction in simplest form. Model the first card or step, then let pairs finish.	Trace it for 24/36. Write each division you make.
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Fraction bubble sort

1. Six students hold fraction cards in a jumbled line (such as 3/4, 1/3, 5/8, 1/2, 2/5, 7/10).
2. Run bubble sort: compare neighbours and swap if the left one is bigger. The class decides each comparison.
3. Count passes and swaps. Ask: 'How did you compare 5/8 and 7/10?' (common denominator or decimals.)
4. Pairs simplify fractions by dividing top and bottom by a common factor until they cannot any more.

Tip: Keep fraction strips at hand for the class to check close comparisons before a swap.

Common mistakes to watch for

- **Comparing only the bottom numbers** or only the top numbers.
- **Stopping bubble sort after one pass** before the list is fully sorted.
- **Simplifying by subtracting** instead of dividing.

Different paces

Support: Sort three fractions with the same denominator first.
Stretch: How many passes does bubble sort need, at most, for 6 items? (5)

Exit check and what to note

Two quick questions at the end of the chapter:

1. Simplify 15/25. (Answer: 3/5)

2. Order 2/5, 1/2 and 3/10 from smallest. (Answer: 3/10, 2/5, 1/2)

Note in the journal: Beginning (needs help to simplify fractions); **Developing** (can simplify fractions with prompts or a partner); **Secure** (can simplify fractions and compare and order fractions alone); **Extending** (can trace a sorting pass and explain why).

Answer key

1. b) 3/4
2. a) 5/8 (25/40 against 24/40)
3. c) 1/2 (3/8, 1/2, 2/3)
4. a) [1/4, 1/2, 3/4]
5. 5/12 (1 - 4/12 - 3/12)
Build it: For example ÷2: 12/18, ÷2: 6/9, ÷3: 2/3 (or ÷12 in one step)
Challenge: 5/6, 7/8, 11/12: each is 1 minus a piece (1/6, 1/8, 1/12); the smaller the missing piece, the bigger the fraction

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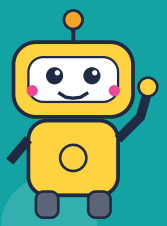
8

TEACHER NOTES · CHAPTER 8 · LEVEL C2

Playing with Constructions

Coder skill: Construction scripts: exact steps and debugging

Goes with NCERT maths, Class 6, chapter 8



At a glance	
Students will	write exact construction steps, use circles to find points and debug a script that goes wrong.
Key words	**construct** (draw exactly with ruler and compass), **arc** (part of a circle), **perpendicular** (at right angles), **debug** (find and fix a mistake)
Materials	Compasses, rulers, set squares, plain paper, script cards
Time	6 periods of 40 minutes
Link	Teach alongside chapter 8 of the NCERT Class 6 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *construct*, *arc*, *perpendicular*, *debug*. Work through the example *A script for the perpendicular bisector of AB* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Construction script swap	Run the activity below. Move between groups and note who can explain the idea.	Write exact construction scripts, follow another pair's script, and debug what went wrong.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Follow this script with a ruler and set square to draw a rectangle 5 cm by 3 cm. Model the first card or step, then let pairs finish.	Draw it in the space. Then write a check step that would catch a mistake.
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Construction script swap

1. Pairs choose a construction: a perpendicular bisector, an equilateral triangle or a 5 cm by 3 cm rectangle.
2. They write the script as numbered, exact steps, without drawing.
3. Swap scripts with another pair, who follow them exactly, step by step.
4. Compare the drawing with the goal. Find and fix the 'bug' in any script that went wrong.

Tip: Ban 'about' and 'some' from scripts. Every length and centre must be exact.

Common mistakes to watch for

- **Changing the compass width** between two arcs that must be equal.
- **Vague steps**, such as 'draw a line here'.
- **Missing a final step**, such as joining the last two points.

Different paces

Support: Follow a ready-made script before writing one.
Stretch: Write a script for a square of side 4 cm using only a compass and ruler (no set square).

Exit check and what to note

Two quick questions at the end of the chapter:

1. Two circles of radius 4 cm have centres 8 cm apart. At how many points do they meet? (Answer: 1 (they just touch))

2. Which tool draws every point that is exactly 5 cm from a point O? (Answer: a compass (it draws a circle))

Note in the journal: Beginning (needs help to write exact construction steps); **Developing** (can write exact construction steps with prompts or a partner); **Secure** (can write exact construction steps and use circles to find points alone); **Extending** (can debug a script that goes wrong and explain why).

Answer key

1. c) 2
2. a) 0 (3 + 3 is less than 8)
3. a) it draws a triangle; it needs: at C draw a right angle, mark CD = 4 cm, join D to A
4. b) circle
5. 5 (4 small and 1 big)
Build it: For example: check DC = 5 cm and that the two diagonals are equal (about 5.8 cm)
Challenge: Draw AB = 5 cm; draw an arc of radius 5 cm with centre A; draw an arc of radius 5 cm with centre B; they cross at C; join AC and BC

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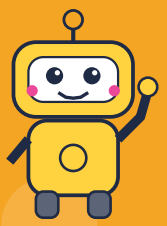
9

TEACHER NOTES · CHAPTER 9 · LEVEL C2

Symmetry

Coder skill: Reflect and rotate: functions that change shapes

Goes with NCERT maths, Class 6, chapter 9



At a glance	
Students will	find lines of symmetry, find the order of rotational symmetry and use a reflect function on a grid.
Key words	**line of symmetry** (fold line where halves match), **reflection** (a mirror image), **rotational symmetry** (matches itself when turned), **order** (how many times it matches)
Materials	Squared paper with a marked mirror line, tracing paper, paper for folding and cutting
Time	6 periods of 40 minutes
Link	Teach alongside chapter 9 of the NCERT Class 6 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *line of symmetry*, *reflection*, *rotational symmetry*, *order*. Work through the example *Reflecting a shape on a grid* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The reflect function	Run the activity below. Move between groups and note who can explain the idea.	Reflect shapes with a reflect function, check by folding, and find rotational symmetry with tracing paper.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Columns on a grid are numbered 1 to 6, and the mirror line is between columns 3 and 4. Model the first card or step, then let pairs finish.	A shape covers columns 1, 2 and 3 of a row. Which columns does its reflection cover? Check that reflect(3) is 4.
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: The reflect function

1. Number the grid columns 1 to 8 with the mirror line between 4 and 5.
2. Students write *reflect(column) = 9 - column* and use it to reflect a shape, square by square.
3. Check by folding along the mirror line or using a mirror.
4. Turn tracing paper over designs to find the order of rotational symmetry.

Tip: Ask 'Why 9?' The two columns next to the mirror (4 and 5) add up to 9, and so does every mirror pair.

Common mistakes to watch for

- **Moving the row** when reflecting in a vertical mirror; only the column changes.
- **Using the wrong constant** in reflect(column) for a different grid.
- **Counting the starting position twice** when finding the order of rotation.

Different paces

Support: Reflect shapes of 2 or 3 squares next to the mirror line.
Stretch: Write reflect(row) for a horizontal mirror line, and reflect a shape in both lines.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Columns are numbered 1 to 8 and the mirror is between 4 and 5. What is reflect(2)? (Answer: 7)

2. What is the order of rotational symmetry of a regular hexagon? (Answer: 6)

Note in the journal: Beginning (needs help to find lines of symmetry); **Developing** (can find lines of symmetry with prompts or a partner); **Secure** (can find lines of symmetry and find the order of rotational symmetry alone); **Extending** (can use a reflect function on a grid and explain why).

Answer key

1. 4
2. c) F
3. c) 3
4. 10
5. b) a diamond-shaped hole
Build it: Columns 6, 5 and 4 (7 - 1, 7 - 2 and 7 - 3)
Challenge: Open answer, for example a rectangle or the letter H

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10

TEACHER NOTES · CHAPTER 10 · LEVEL C2

The Other Side of Zero

Coder skill: Integers in data: lowest, highest and range

Goes with NCERT maths, Class 6, chapter 10



At a glance	
Students will	compare and order integers, add and subtract with negatives and find the range of data with negatives.
Key words	**integer** (a whole number or its negative), **negative** (less than zero), **range** (highest minus lowest), **minimum** (the lowest value)
Materials	A vertical number line from -5 to 5 (a 'lift' on the wall), temperature data cards, thermometer picture
Time	6 periods of 40 minutes
Link	Teach alongside chapter 10 of the NCERT Class 6 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *integer*, *negative*, *range*, *minimum*. Work through the example *Tracing a lift* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The lift operator	Run the activity below. Move between groups and note who can explain the idea.	Trace lift programs on a vertical number line and find the range of temperature data with negatives.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A temperature log for one week in Leh in winter (°C). Model the first card or step, then let pairs finish.	Run it on the week: -3, -7, 2, -1, -8, 0, 4. What are the lowest, the highest and the range?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: The lift operator

1. Draw floors -5 to 5 on the wall; a student is the lift, starting at 0.
2. Read lift programs: *up 4, down 6, up 1*. The class predicts each floor before the lift moves.
3. Groups get a week of winter temperatures with negatives and find the lowest, highest and range.
4. Ask: 'Why is the range 4 - (-8) and not 4 - 8?'

Tip: Use the lift to show subtracting a negative: going from -8 up to 4 is 12 floors.

Common mistakes to watch for

- **Thinking -7 is bigger than -3** because 7 is bigger than 3.
- **Range with negatives**: highest - lowest, so 4 - (-8) = 12.
- **Moving the wrong way** for 'down'.

Different paces

Support: Use floors 0 to 5 first, then add the basements.
Stretch: Write a program that finds the lowest and highest in a temperature list in one pass, and trace it.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Start at -3. Go up 7, then down 6. Where are you? (Answer: -2)

2. Temperatures are 4, -2, 7 and -5. What is the range? (Answer: 12)

Note in the journal: Beginning (needs help to compare and order integers); **Developing** (can compare and order integers with prompts or a partner); **Secure** (can compare and order integers and add and subtract with negatives alone); **Extending** (can find the range of data with negatives and explain why).

Answer key

1. a) -7
2. -5 °C
3. c) W (T18, U10, V2, W-6)
4. -5
5. -1
Build it: Lowest -8, highest 4, range 4 - (-8) = 12 degrees
Challenge: -45 m (-120 + 45 - 30 + 60), which is 45 m below sea level

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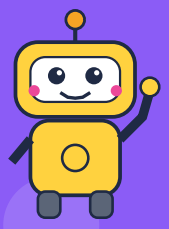
AI1

TEACHER NOTES · AI CHAPTER 1 · LEVEL C3

Introduction to AI

AI skill: Tell AI from automation; how machines learn

Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 1



At a glance	
Students will	explain what AI is, tell AI from automation and name three ways machines learn.
Key words	**AI** (machines that learn and decide), **automation** (fixed steps, same result), **labelled data** (examples with the right answer), **model** (what an AI learns)
Materials	'Is it AI?' machine cards, a small hidden object, sorting boxes labelled AI and Automation
Time	7 periods of 40 minutes
Link	Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 1.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *AI*, *automation*, *labelled data*, *model*. Work through the example *AI or automation?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Is it AI?	Run the activity below. Move between groups and note who can explain the idea.	Sort machines into AI and automation with reasons, and act out learning by reward.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a sorting rule for 'AI or automation?' as code cards. Model the first card or step, then let pairs finish.	Use your rule on: a calculator, a spam filter, an alarm clock, a photo app that finds faces, a music app that suggests songs, street lights on a timer.
5	Three ways to learn	Act out supervised (cards with answers), unsupervised (group cards with no labels) and reinforcement (hot and cold) learning.	Name the type of learning in each example on the board.
6	AI around us	Pairs list 5 machines at home or school and classify them; share the hardest one.	Classify everyday machines and justify each choice.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Is it AI?

1. Groups sort machine cards (microwave timer, spam filter, face unlock, street lights on a timer, voice assistant) into AI and Automation.
2. Each group explains one decision using the test: does it learn from data, or always do the same thing?
3. Play 'hot and cold': one student finds a hidden object guided only by claps that get louder when closer.
4. Ask: 'Which kind of learning was that?' (learning by reward: reinforcement learning).

Tip: Keep one card that is hard to decide (a washing machine with sensors). Debate is the point.

Common mistakes to watch for

- **Thinking anything electronic is AI.** Ask: does it learn and improve?
- **Mixing up supervised and unsupervised learning.** Supervised learning has labels (answers).
- **Thinking AI is always right** or that it thinks like a person.

Different paces

Support: Sort only 4 clear cards first.
Stretch: Find a machine that could be made 'smart' with AI, and name the data it would need.

If you have computers (optional)

Show a classroom-safe image classifier demo trained on two classes (for example two kinds of stationery). Ask students to predict when it will make mistakes, then test.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Face unlock on a phone: AI or automation? (Answer: AI: it learnt your face from pictures)

2. Learning from examples that come with the right answers is called ... (Answer: supervised learning)

Note in the journal: Beginning (needs help to explain what AI is); **Developing** (can explain what AI is with prompts or a partner); **Secure** (can explain what AI is and tell AI from automation alone); **Extending** (can name three ways machines learn and explain why).

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Answer key

1. c) a fixed-time traffic signal
2. a) labelled data
3. c) reinforcement learning
4. b) unsupervised learning
5. b) 1956, at Dartmouth College
Build it: AI: spam filter, face-finding photo app, song suggestions. Automation: calculator, alarm clock, timer street lights
Challenge: Open answer, for example a fan that learns when the family is usually home, using data about times and temperatures

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AI2

TEACHER NOTES · AI CHAPTER 2 · LEVEL C3

Basic Data Concepts

AI skill: Types of data; collect, organise and show it

Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 2



At a glance	
Students will	name types of data, collect and organise data and choose a table or chart to show data.
Key words	**data** (raw facts), **information** (data organised to be useful), **survey** (asking people questions), **classify** (group similar items)
Materials	Survey sheets, tally charts, squared paper, sorting cards showing different kinds of data
Time	7 periods of 40 minutes
Link	Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 2.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *data*, *information*, *survey*, *classify*. Work through the example *The eco club's tree-planting drive* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: From question to chart	Run the activity below. Move between groups and note who can explain the idea.	Collect survey data, organise it in a table, chart it and write what it shows.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Plan and run a small survey: *How do you come to school?* Model the first card or step, then let pairs finish.	Ask 10 classmates. Draw your tally table and bar chart in the space, then write your conclusion.
5	Data types hunt	Show examples (a voice note, a marks table, a photo, a temperature log); groups sort them into number, text, image and sound.	Sort examples into data types and explain one choice.
6	Is the data fair?	Discuss the 'cricket team survey' challenge: who was asked, and who was missing?	Spot a biased sample and suggest a fairer way to collect.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: From question to chart

1. The class agrees one survey question, such as 'How do you come to school?'.
2. Groups collect answers with tally marks, then organise them in a frequency table.
3. Each group draws a bar chart with a title, labelled axes and a scale.
4. Groups write one piece of information the chart shows, and one question it cannot answer.

Tip: Ask 'Is this data or information?' at each stage. Raw tallies become information once organised.

Common mistakes to watch for

- **Leading questions** that push people towards one answer.
- **Counting someone twice** or leaving people out.
- **Charts without a scale or labels.**

Different paces

Support: Give a ready-made frequency table to chart.
Stretch: Compare two classes' charts and write one difference that the data explains, and one it cannot.

If you have computers (optional)

Enter the survey data in a spreadsheet and let students watch the chart update as values change.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Measuring a plant's height every week collects which type of data? (Answer: number data)

2. Putting similar data into groups is called ... (Answer: classifying)

Note in the journal: Beginning (needs help to name types of data); **Developing** (can name types of data with prompts or a partner); **Secure** (can name types of data and collect and organise data alone); **Extending** (can choose a table or chart to show data and explain why).

Answer key

1. d) sound
2. a) number
3. 20 (35 - 15)
4. b) classifying
5. a) a bar chart
Build it: Open answer; check that the tallies add up to 10 and that the chart has a title, labels and a scale
Challenge: No: 5 people is a small sample, and they all play cricket, so the data is biased

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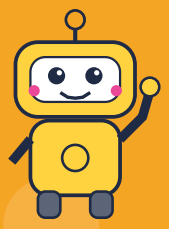
AI3

TEACHER NOTES · AI CHAPTER 3 · LEVEL C3

Patterns and Decisions

AI skill: Learn a rule from labelled examples, then decide

Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 3



At a glance	
Students will	tell an observation from a conclusion, find a rule from labelled examples and explain why more and fairer data helps.
Key words	**pattern** (something that repeats), **observation** (what the data shows), **conclusion** (what the data means), **prediction** (a best guess for a new case)
Materials	AI picture cards: 12 Zib and Zob creature cards (8 labelled for training, 4 unlabelled for testing), scoring slips
Time	7 periods of 40 minutes
Link	Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 3.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *pattern*, *observation*, *conclusion*, *prediction*. Work through the example *Learning when a mango is ripe* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Train the classifier	Run the activity below. Move between groups and note who can explain the idea.	Find a rule from labelled examples, test it on unseen cards, and work out how often it was right.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Be the classifier. Find the rule that tells a **Zib** from a **Zob**. Model the first card or step, then let pairs finish.	Fill in the rule, then label the three new creatures. Does colour matter?
5	Testing and accuracy	Order jumbled step cards: collect examples, find the pattern, make a rule, test, improve. Work out accuracy as a fraction.	Order the steps of training a model and calculate accuracy, such as 3 of 4.
6	Observation or conclusion?	Read statements about weather and practice data; students sort them into observations and conclusions.	Sort statements and explain the difference, with an example of each.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Train the classifier

1. Show the 8 labelled training cards. Groups list features (colour, stripes, spots) and predict a rule.
2. Ask: 'Does colour matter? How do you know?' Groups drop features that do not fit every card.
3. Hand out the 4 test cards. Groups classify them, then you reveal the labels; groups count how many were right.
4. Show a creature with both stripes and spots. Groups decide what the classifier should say.

Tip: Colours are mixed on purpose, so the only real pattern is stripes (Zib) or spots (Zob).

Common mistakes to watch for

- **A rule from one card.** 'Orange ones are Zibs' fits one card but not the others.
- **Using the test cards to make the rule**, so the score is not a fair test.
- **Thinking AI is always right.** A rule from a few examples can fail on something new.

Different paces

Support: Start with 4 training cards that differ only in pattern.
Stretch: Add a creature with a new feature (horns): must the rule change? Write the score as a fraction.

If you have computers (optional)

Train a two-class image model on the creature cards with a no-code tool, then test it on the unlabelled cards and on a creature with both patterns.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Why do we test on cards the classifier has not seen? (Answer: to check it learnt the pattern, not the answers)

2. A classifier got 9 of 12 test cards right. What fraction is that in simplest form? (Answer: 3/4)

Note in the journal: Beginning (needs help to tell an observation from a conclusion); **Developing** (can tell an observation from a conclusion with prompts or a partner); **Secure** (can tell an observation from a conclusion and find a rule from labelled examples alone); **Extending** (can explain why more and fairer data helps and explain why).

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Answer key

1. a) an observation
2. b) a conclusion
3. a) spam (it has the words 'win', 'free' and 'click')
4. a) it may not recognise it: its training data was not varied enough
5. a) practise daily
Build it: If it has stripes, predict Zib, else Zob. New creatures: Zib, Zob, Zib. Colour does not matter: each colour appears in both groups
Challenge: It cannot be sure, because it never saw one like this; it should say 'not sure', and we should add labelled examples like it to the training data

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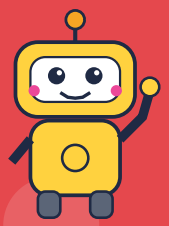
AI4

TEACHER NOTES · AI CHAPTER 4 · LEVEL C3

Ethics and Digital Responsibility

AI skill: Strong passwords, privacy and digital footprints

Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 4



At a glance	
Students will	make and protect a strong password, spot phishing and explain active and passive footprints.
Key words	**digital footprint** (the trail you leave online), **phishing** (tricking people to get secrets), **plagiarism** (copying work as your own), **privacy** (control of your personal data)
Materials	Scenario cards (phishing email, stranger in a game chat, copied homework), password strength cards, chart paper
Time	7 periods of 40 minutes
Link	Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 4.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *digital footprint*, *phishing*, *plagiarism*, *privacy*. Work through the example *Make a strong password you can remember* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Password lab and scenario cards	Run the activity below. Move between groups and note who can explain the idea.	Make example strong passphrases, decide what to do in online scenarios, and write a safety rule.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A safety rule for messages, written as code cards. Model the first card or step, then let pairs finish.	Write your own rule for this: a stranger in a game chat asks for the name of your school.
5	Digital footprint map	Students list things they do online in a week and mark each as an active or passive footprint.	Map their own digital footprint, active and passive.
6	Copying and credit	Show a paragraph copied from a website and the same idea in the student's own words with a source. Discuss plagiarism.	Rewrite a short paragraph in their own words and add where it came from.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Password lab and scenario cards

1. Show weak passwords (riya2014, 123456) and ask why they are weak.
2. Students build a passphrase from three unrelated words plus a number and a symbol, but do not write real ones.
3. Groups draw scenario cards and decide: safe or unsafe, and what to do (do not click, tell an adult, report).
4. Each group writes one 'safe online' rule as if-then code cards.

Tip: Never ask students to share or write their real passwords. Use made-up examples only.

Common mistakes to watch for

- **Thinking a long password is enough** even if it is a name or birthday.
- **Sharing passwords with friends.** Passwords are never shared.
- **Thinking a passive footprint is harmless** because they did not post it.

Different paces

Support: Use the strength checklist: long, mixed letters, a number, a symbol, not personal.
Stretch: Find three clues that an email is phishing, using a made-up example.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Which is stronger: Riya2014 or Kite!Mango7River? (Answer: Kite!Mango7River)

2. An email asks you to reply with your password to 'unlock' your account. What is this called? (Answer: phishing)

Note in the journal: Beginning (needs help to make and protect a strong password); **Developing** (can make and protect a strong password with prompts or a partner); **Secure** (can make and protect a strong password and spot phishing alone); **Extending** (can explain active and passive footprints and explain why).

Answer key

1. c) Kite!Mango7River
2. a) phishing
3. a) active
4. b) passive
5. a) plagiarism
Build it: For example: if a stranger asks where I live or study, I do not answer, I block or report them, and I tell a trusted adult
Challenge: Open answer, for example active: posts, comments, online forms; passive: location history, cookies, suggestions based on your likes

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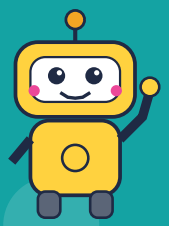
P1

TEACHER GUIDE · PROJECT 1 · 20 HOURS · LEVEL C3

Weather Across India

Project skill: Collect, organise and present real data

CBSE: 40 project hours a year in Class 6, as two projects of 20 hours



At a glance	
Big question	How different is the weather in five Indian cities, and what patterns can we find?
Students will	collect two weeks of real weather data for five cities, organise and chart it, find patterns and present conclusions with their limits
Groups	Teams of 3 or 4, with roles that rotate at the halfway point
Materials	Newspapers or printed city forecasts, data sheet pads (kit), squared paper, chart paper, the student book's project pages
Time	30 periods of 40 minutes (about 2 periods a week for 10 weeks)
Evidence	Data sheets with sources, calculations of highest, lowest and range, two charts, a poster or talk, and the completed rubric

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Week by week

Weeks	Stage	Students do	Teacher does
1	Define	Agree the question, choose five cities and decide what to record each day.	Form teams and give roles. Help each team choose five cities in different parts of India and agree what to record.
2-3	Collect	Record the highest and lowest temperature and rain for each city every day for two weeks. Write the source.	Show how to read the weather table in a newspaper or an official forecast. Check every team's sheet and sources daily in the first week.
4	Organise	Put the data in one table per city. Look for missing or odd values and check them.	Model one city's table. Ask teams to look for gaps and odd values, such as a typing error of 350 °C.
5	Calculate	For each city find the highest, the lowest and the range. Count the rainy days.	Revise highest, lowest and range with negative numbers (chapter 10). Check one calculation per team.
6	Show	Draw a bar chart of the ranges, and a bar chart of one city's daily highs.	Check that chart scales are even and labelled, and that bars below 0 are shown correctly.
7	Find patterns	Write three observations and one conclusion. Is two weeks of data enough to be sure?	Push for 'because' in every conclusion, and ask: 'Is two weeks enough to be sure?'
8	Predict and test	Predict tomorrow's highest temperature for one city, then check the real value.	Discuss why predictions can be wrong, and link to how AI makes predictions from past data.
9	Present	Make a poster or a 3-minute talk: question, data, charts, findings and limits.	Run 3-minute talks. Each listener gives 'two stars and a wish' as feedback.
10	Reflect	Score your work with the rubric and write what you would improve.	Students score themselves with the rubric; you score too. Talk about any row where the scores differ.

Setting up

- ✓ Agree one time of day and one source type for all readings (for example the next morning's newspaper).
- ✓ Agree units (°C) and how to write negatives before collecting.
- ✓ Any internet use is with an adult and on trusted sites only.
- ✓ Keep a 'source' column on every data sheet from day one.

Team roles

Give each student a role and rotate roles at the halfway point, so everyone collects, checks, builds and presents.

Collector: records each day's values and the source.

Checker: checks values for gaps, typing errors and odd numbers.

Chart maker: plans and draws the charts with honest scales.

Presenter: leads the talk and the poster.

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Common pitfalls

- **Mixing forecasts and actual readings** on different days. Pick one and stick to it.
- **Range errors with negatives**: 2 - (-4) is 6, not -2.
- **Conclusions that go beyond the data**, such as 'Leh is always below freezing' from one fortnight.

Different paces

Support: Use three cities and one week of data, with a partly filled data sheet.
Stretch: Add rainfall and compare: do cities with more rainy days have a smaller temperature range?

If you have computers (optional)

Enter the data in a spreadsheet, use formulas for highest, lowest and range, and compare the charts with the hand-drawn ones.

Marking with the rubric

Score each row from 1 to 4 using the evidence in the team's sheets, charts and talk; the total is out of 24. Most teams score 14 to 20. Agree the scores with students and record the total in the journal and on the term summary.

What we check	What excellent (4) looks like
Question	Clear, and can be answered with the data collected.
Data	Complete, from a named source, with odd values checked.
Thinking	Highest, lowest and range are right; patterns are spotted.
Charts and talk	Charts have titles, scales and labels; the talk is clear.
Honesty and limits	Says what the data cannot show; nothing copied without credit.
Teamwork	Everyone has a job and the work is shared fairly.

What a strong project sounds like

'Over 14 days, Chennai's temperature range was only 4 degrees, but Leh's was 11. Coastal cities seemed to change less. We used the same newspaper every day. Two weeks is not enough to be sure, so we would like to collect data for a whole year.'

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P2

TEACHER GUIDE · PROJECT 2 · 20 HOURS · LEVEL C3

A Smarter School Routine

Project skill: Design, test and improve an algorithm

CBSE: 40 project hours a year in Class 6, as two projects of 20 hours



At a glance	
Big question	Can we design a better routine for a real problem at school, and prove it works?
Students will	choose a real school problem, measure it, design a routine as an algorithm and flowchart, try it, and compare before and after data
Groups	Teams of 3 or 4, with roles that rotate at the halfway point
Materials	Stopwatch or wall clock, data sheet pads, flowchart stencil cards (kit), chart paper, the student book's project pages
Time	30 periods of 40 minutes (about 2 periods a week for 10 weeks)
Evidence	Before and after data, the flowchart and its trace by a friend, a comparison chart, the presentation and the completed rubric

Week by week

Weeks	Stage	Students do	Teacher does
1	Define	Choose one problem. Write who it affects and how you will measure it.	Help teams choose a small, real problem they can measure, such as a queue or a messy shelf. Agree it with the staff involved.
2	Observe	Collect 'before' data: for example, time five lunch breaks at the water cooler.	Check that teams measure the same way each time: same place, same time of day, same start and stop.
3-4	Design	Break the problem down. Write your new routine as steps, then draw it as a flowchart.	Model a simple flowchart. Check every decision diamond has both a yes and a no arrow.
5	Check	Trace the flowchart with a friend. Fix any step that is unclear or missing.	Pair teams to trace each other's flowcharts aloud and note unclear steps.
6-7	Try it	With your teacher's permission, use the routine for a week and collect 'after' data.	Get permission from the staff involved before the trial, and check data collection each day.
8	Compare	Put the before and after data in a table and a bar chart.	Help teams put before and after values side by side and draw a fair comparison chart.
9	Present	Show the problem, the flowchart, the data and the result to your class.	Invite the staff member involved to the presentations if possible.
10	Reflect	Score your work with the rubric. Was the routine fair to everyone?	Students score themselves with the rubric; you score too. Talk about fairness: did the routine work for everyone?

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Setting up

- Choose problems the team can measure and that staff agree may be tried.
- Collect counts and times only: no names or personal details.
- Agree how long the trial will run (about a week) before it starts.
- Keep the flowchart shapes standard: ovals, boxes and diamonds.

Team roles

Give each student a role and rotate roles at the halfway point, so everyone collects, checks, builds and presents.

Timekeeper: measures and records before and after data.

Designer: drafts the algorithm and the flowchart.

Tester: traces the flowchart and finds unclear or missing steps.

Presenter: leads the comparison and the talk.

Common pitfalls

- **Measuring before and after in different ways**, so the comparison is not fair.
- **Decisions with only one arrow** in the flowchart.
- **Trying a routine without permission** or without telling the people affected.

Different paces

Support: Give a partly drawn flowchart for a familiar routine (the water cooler) to complete.
Stretch: Try two different routines on different days and recommend one, using the data.

If you have computers (optional)

Draw the flowchart in a free online diagram tool or a slides app and present it on the projector.

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Marking with the rubric

Score each row from 1 to 4 using the before and after data, the flowchart, the trace and the talk; the total is out of 24. Give credit for honest reporting even when the routine did not improve things.

What we check	What excellent (4) looks like
Problem	A real, specific problem with a clear way to measure it.
Algorithm	Steps are exact, in order, and cover what to do when something goes wrong.
Flowchart	Correct shapes; every decision has a 'yes' and a 'no' path.
Testing	Before and after data are collected the same way and compared.
Fairness	The routine works for everyone, including younger children.
Talk and teamwork	A clear talk with the flowchart and chart; the work is shared fairly.

What a strong project sounds like

'Before, 20 children took 9 minutes to fill their bottles. With our routine, it took 6 minutes on average over five days. Younger children found the yellow line confusing, so we added a picture sign. We would test it for another week to be sure.'

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Observation journal

Class 6 ____ Section ____ Term ____ Chapters _____. Circle one level per skill: **B**eginning, **D**eveloping, **S**ecure, **E**xtending. You may photocopy this page.

Student	Algorithms	Data	AI ideas	Projects	Notes
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
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	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	

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Term summary for the report card

One copy per child each term. Fill it from the observation journal and the child's self-checks. You may photocopy this page.

Name _____ Class _____ Section _____ Term _____

Area	Level (circle one)	Teacher comment
Computational thinking (overall)	B D S E	
Algorithms and functions	B D S E	
Data and charts	B D S E	
AI ideas and digital responsibility	B D S E	
Projects (rubric score)	B D S E	

Writing comments

Write one strength and one next step, in words a parent understands. For example: *Explains steps clearly and fixes bugs in routes; next step: spotting the core of a pattern.*

Class teacher _____ Date _____

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